Project D.E.P.T.H. (Dynamic Environmental Profiling and Threat-Identification for Hypoxia): Design and Evaluation of an Autonomous Surface Vehicle for Spatiotemporal Water Quality Mapping in Nile Tilapia (*Oreochromis niloticus*) Aquaculture

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This study presents the design and evaluation of Project D.E.P.T.H. (Dynamic Environmental Profiling and Threat-Identification for Hypoxia), an autonomous surface vehicle (ASV) developed to address the persistent problem of sudden fish kills caused by hypoxia in Nile Tilapia (*Oreochromis niloticus*) aquaculture ponds. Traditional single-point, manual water quality monitoring methods fail to detect localized dissolved oxygen (DO) "dead zones" that can emerge and expand before farmers are able to respond, often resulting in catastrophic stock losses. The system integrates an ESP32-S3 microcontroller with pH, temperature, and DO sensors on a catamaran hull, navigating a predefined 16-point grid using dead reckoning. Collected data are transmitted to a Firebase-backed web dashboard that generates interpolated water quality heatmaps and actionable management alerts. The study was conducted across three tilapia pond deployments using a Method Comparison Study design. Navigational performance was assessed using Mean Absolute Terminal Error (MATE), sensor accuracy was evaluated via Percent Error, RMSE, and paired t-tests, operational reliability was measured by Coefficient of Variation (CV), spatial mapping accuracy was validated through ground truth comparison, and user interface accessibility was assessed with the System Usability Scale (SUS). Results showed that the robot achieved a MATE well below the 3.0-meter threshold in all trials; the pH and temperature sensors demonstrated high accuracy (Mean PE < 0.56% and < 0.51%, respectively); and the DO sensor performed adequately for hypoxia detection despite higher variability in outdoor deployments. Mission time CV was 2.94%, indicating high temporal reliability, while terminal ATE CV was 50.67%, reflecting cross-pond environmental variability. Heatmap interpolation produced MAE values ranging from 0.1123 to 0.1885 across all parameters. The web dashboard achieved a mean SUS score of 79.17, classified as Acceptable. These findings demonstrate that Project D.E.P.T.H. is a viable solution for spatiotemporal water quality monitoring in small-scale tilapia aquaculture, with strong potential to reduce hypoxia-induced fish kills through early detection and proactive pond management.

Keywords: *aquaculture monitoring, autonomous surface vehicle, dead reckoning, dissolved oxygen, hypoxia, Internet of Things, Nile Tilapia, spatiotemporal mapping, water quality*

INTRODUCTION

Aquaculture is a cornerstone of the Philippine economy and a vital source of dietary protein. Nile Tilapia (*Oreochromis niloticus*) farming in the Philippines is one of its most important components. According to the Food and Agriculture Organization of the United Nations (FAO, n.d.), tilapia stands as the second-most consumed fish in the Philippines and offers an accessible and affordable protein source for millions of Filipinos, with average annual consumption reaching approximately 4.7 kg per person. It is notably cheaper than pork or chicken, having an average retail price of PhP 167.62 (PSA, 2024), and contributes approximately 12% of the country's animal protein intake, directly impacting national food security and economic stability.

However, tilapia production is frequently threatened by sudden fish kills caused by poor water quality. Crucial parameters including water temperature, pH, and DO levels must be carefully managed, as deviations outside optimal ranges can induce stress, suppress immune function, and cause direct mortality (Hemal et al., 2024). When DO concentrations fall below 3 mg/L, fish experience significant stress, and levels between 0.7 to 1.6 mg/L have been empirically shown to induce severe distress and mortality in tilapia (Li et al., 2018). A significant concern is the rapid development of localized "dead zones" where hypoxia becomes acute, often causing catastrophic loss of stock and severe economic damage to farmers (Burtle, 2014).

The primary challenge in preventing these issues lies in the method of monitoring. Traditional methods typically involve time-consuming and labor-intensive processes of collecting spot water samples and analyzing them using handheld field meters (Madrid & Zayas, 2007). The fundamental flaw in this approach lies in its intermittent nature and single-point data acquisition, which provides only a small snapshot of pond conditions. A single favorable reading can falsely reassure a farmer, allowing a nearby "dead zone" to expand unnoticed (Vasudevan & Baskaran, 2021). Modern aquaculture has increasingly adopted Internet-of-Things (IoT) systems to monitor water quality continuously (Flores-Iwasaki et al., 2025); however, existing systems still fail to account for the spatial heterogeneity of water quality within a pond. Autonomous surface vehicles (ASVs) for water quality monitoring have been developed by various researchers, but many rely on expensive, specialized sensors and custom-made hardware, making them inaccessible to fish farmers (Li et al., 2020; Ferri et al., 2014).

This project addresses that critical gap by developing Project D.E.P.T.H., an autonomous surface vehicle that continuously and systematically maps water quality across a pond. The system integrates onboard sensors (pH, temperature, and DO) with navigation and data logging to deliver real-time, spatiotemporal water quality profiles. The data is presented on a web-based dashboard using JavaScript and Google Firebase for storage and visualization. This study aimed to: (1) evaluate the navigational performance, sensor accuracy, and operational reliability of the Project D.E.P.T.H. robot; (2) determine whether significant differences exist between the robot's sensor readings and those of calibrated commercial sensors; and (3) assess the effectiveness of the web-based dashboard in terms of visual interpolation accuracy and user experience.

MATERIALS AND METHODS

Research Design

The experimental research design employed was a Method Comparison Study. This quantitative approach was specifically chosen to rigorously evaluate the effectiveness of the Project D.E.P.T.H. robot's sensors by comparing their performance against a validated, "gold standard" instrument. The core methodology involved collecting paired data sets, where measurements of water parameters—specifically pH, temperature, and DO—were taken simultaneously by the autonomous surface robot and a calibrated commercial handheld meter at the same location. Statistical tools including Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) were used to quantitatively determine whether the ASV could provide data comparable to that of a professional-grade device.

Research Setting

The study was conducted in Barangay Gracia, Zone 6, Tagoloan, Misamis Oriental, Philippines. Field trials were performed in three small-sized freshwater aquaculture ponds owned by three consenting tilapia farmers (Farmer A, Farmer B, and Farmer C), situated in close geographic proximity to one another to minimize logistical variability across trials.

Respondents and Sampling

The respondents consisted of three (3) Nile Tilapia (Oreochromis niloticus) fish farm owners who served a dual role: as the owners of the tilapia ponds in which the ASV was deployed for field testing, and as the end-user evaluators of the web-based water quality dashboard through the System Usability Scale (SUS). The study employed purposive sampling, a non-probability technique in which participants were selected based on specific, pre-determined criteria: operational tilapia ponds of suitable dimensions for ASV deployment, geographic proximity to minimize logistical variability, and willingness to provide formal consent for the deployment of research equipment.

Hardware Development

The Project D.E.P.T.H. ASV was built on a catamaran hull using two 3.5-inch outer diameter Schedule 40 PVC pipes as pontoons, with a 6 mm marine-grade plywood platform sealed with epoxy resin. Locomotion was provided by two 12V bilge-pump motors driven by electronic speed controllers (ESC), enabling differential steering. Navigation was achieved through dead reckoning, wherein the ESP32-S3 microcontroller maintained an internal odometry model that tracked cumulative motor runtime and turning angles to compute the robot's position relative to a pre-programmed 16-waypoint grid (Siegwart et al., 2011).

The sensor system included a pH sensor probe (SEN0161), a DS18B20 temperature sensor, and a DFRobot Gravity Analog Dissolved Oxygen Sensor. All electronic components were housed in a waterproof plastic enclosure measuring 35 × 24 × 19 cm. Power was supplied by a 2250 mAh 11.1V Li-Po battery, with a DC-DC buck converter stepping voltage down to 5V for the electronic components.

Software Development

Custom firmware was developed in the Arduino-ESP32 environment. At startup, the ESP32-S3 initialized its Wi-Fi interface, synchronized the internal clock via NTP, and loaded the predefined 16-point waypoint grid. At each waypoint, the firmware simultaneously read all three sensors. The logged data—comprising timestamp, Point ID, pH, temperature (°C), and DO (mg/L)—were stored internally. Upon completing the mission, the ESP32 transmitted the data to a Google Firebase Realtime Database via HTTP POST requests.

The web-based dashboard was built using HTML and the Firebase SDK, enabling real-time data synchronization. The dashboard applied bilinear interpolation to the 16 discrete waypoint readings to generate continuous spatiotemporal heatmaps of pH, temperature, and DO across the pond surface. A logic-gate analysis was simultaneously executed: if any parameter fell outside defined safe thresholds—such as DO dropping below 3 mg/L (the critical hypoxia level for Nile Tilapia)—the system automatically generated actionable management alerts.

Data Collection

Project D.E.P.T.H. was deployed in three local tilapia ponds following formal partnership agreements with the consenting farm owners. During each autonomous mission, the robot navigated through 16 waypoints while collecting simultaneous sensor readings of pH, DO, and temperature alongside readings from a calibrated commercial handheld meter. The Absolute Terminal Error (ATE) was recorded at the conclusion of each mission by measuring the physical Euclidean distance between the vehicle's terminal position and the target marker. Following the field trials, a "ground truth" validation process was initiated: the researchers selected 13 random coordinates per pond not directly traversed by the robot, took manual measurements at these locations, and compared them against the dashboard's interpolated values (Gomez et al., 2021). Finally, the three participating tilapia farmers completed a standardized 10-item SUS questionnaire after interacting with the web-based platform (Brooke, 1996).

Data Analysis

Navigational accuracy was quantified using the Absolute Terminal Error (ATE) and Mean Absolute Terminal Error (MATE), calculated as the Euclidean distance between the robot's intended stop point and its actual physical location. Sensor accuracy was assessed using Percent Error (PE) and RMSE, comparing robot readings against the calibrated commercial reference. A paired t-test ($df = 15$, $\alpha = 0.05$) was conducted to determine whether differences between the robot and commercial sensor readings were statistically significant (Ibrahim et al., 2016). Operational reliability was evaluated using the Coefficient of Variation (CV) applied to the ATE and mission completion time across the three trials; a CV < 10% was the benchmark for high reliability (Afshar-Mohajer et al., 2017). Spatial mapping accuracy was validated through Absolute Error (AE), MAE, and RMSE comparing heatmap-interpolated values against manual ground truth measurements. User experience was scored using the SUS formula, with a mean score ≥ 68 indicating "Acceptable Usability" (Brooke, 1996).

RESULTS AND DISCUSSIONS

Navigational Performance

The navigational performance of the dead reckoning system was evaluated by computing the ATE at each of the 16 predefined waypoints across three pond deployments. Table 1 summarizes the results.

Table 1. *Summary of Navigational Performance per Pond*

Pond	MATE	RMSE	SD	Max	Min
Pond 1	**2.0187**	**2.2982**	**1.1344**	**3.90**	**0.20**
Pond 2	**3.1688**	**3.4635**	**1.4439**	**5.00**	**0.50**
Pond 3	**2.3375**	**2.6249**	**1.2334**	**4.20**	**0.30**

Note. MATE = Mean Absolute Terminal Error; RMSE = Root Mean Square Error; SD = Standard Deviation. All values are in centimeters (cm).

Positional errors increased progressively with each successive waypoint in all three pond trials, which is the expected behavior of a dead reckoning system operating without external correction—each new waypoint is calculated from the robot's estimated, not actual, position, causing positional error to accumulate (Siegwart et al., 2011). Pond 1 recorded the lowest error profile (MATE = 2.02 cm), consistent with it being the most sheltered deployment environment. Pond 2 exhibited the highest errors (MATE = 3.17 cm), attributable to greater surface wind exposure and more active water movement—a "crabbing" effect where sustained lateral forces from wind offset the robot's heading (Borenstein & Feng, 1996). Despite the known accumulation mechanism, the overall mean MATE across all three trials was 0.0251 m, substantially below the 0.5-centimeter threshold in Hypothesis H01. These results are consistent with Davis et al. (2023), who demonstrated that timed open-loop path-following can achieve sufficient navigational fidelity for grid-based aquaculture sampling in small enclosed water bodies. H01 was accepted: dead reckoning is a viable and cost-effective navigation strategy for small-scale tilapia pond monitoring.

Sensor Accuracy

The accuracy of the pH, temperature, and DO sensors was assessed by comparing their simultaneous readings with those of a calibrated commercial handheld meter at 16 sampling points per trial. Tables 2, 3, and 4 summarize the results.

Table 2. *Summary of pH Sensor Accuracy Across All Trials*

Pond	Mean PE (%)	RMSE	Mean Diff	t-statistic
Pond 1	0.4872	0.0370	-0.0356	-13.82*
Pond 2	0.5248	0.0406	-0.0400	-21.91*
Pond 3	0.5539	0.0406	-0.0400	-21.91*

Note. PE = Percent Error; RMSE = Root Mean Square Error; Mean Diff = mean difference between robot and manual readings; n = 16 per pond; α = 0.05. * = statistically significant.

Table 3. *Summary of Temperature Sensor Accuracy Across All Trials*

Pond	**Mean PE (%)**	**RMSE (°C)**	**Mean Diff**	**t-statistic**
Pond 1	0.2004	0.0707	-0.0500	-3.87*
Pond 2	0.5036	0.1532	0.0169	0.43
Pond 3	0.4750	0.1479	0.0138	0.36

Note. PE = Percent Error; RMSE = Root Mean Square Error (°C); * = statistically significant.

Table 4. *Summary of DO Sensor Accuracy Across All Trials*

Pond	**Mean PE (%)**	**RMSE (mg/L)**	**Mean Diff**	**t-statistic**
Pond 1	0.6769	0.0354	-0.0081	-0.91
Pond 2	3.0751	0.1756	0.1031	2.81*
Pond 3	3.0608	0.1492	-0.0150	-0.39

Note. PE = Percent Error; RMSE = Root Mean Square Error (mg/L); * = statistically significant.

The pH sensor demonstrated consistently high accuracy across all three ponds. Mean Percent Errors ranged from 0.4872% to 0.5539%, considerably below the commonly accepted 5% threshold for low-cost environmental sensors (Bouillod et al., 2022). The negative mean differences indicate a systematic underestimation that is a well-documented characteristic of analog electrochemical pH probes and can be corrected through a simple linear recalibration factor (Mohd Jais et al., 2024). The high t-statistics confirm this systematic bias is statistically detectable, though its absolute magnitude is operationally negligible, consistent with findings by Hemal et al. (2024) and Zlatev et al. (2024).

The DS18B20 temperature sensor also performed with high accuracy, with Mean PE values below 0.51% across all ponds. Higher RMSE values in Ponds 2 and 3 (0.1532°C and 0.1479°C, respectively) are attributable to localized thermal stratification in the outdoor ponds due to differential solar heating and evaporative cooling (Pedrazzani et al., 2020). The non-significant t-statistics for Ponds 2 and 3 indicate no consistent directional bias, confirming measurement noise rather than systematic instrument drift (Afshar-Mohajer et al., 2017), consistent with Chang et al. (2021).

The DO sensor exhibited a more variable accuracy profile. Pond 1 recorded excellent accuracy (Mean PE = 0.6769%, RMSE = 0.0354 mg/L), while Ponds 2 and 3 showed higher Mean PE values of approximately 3.07%, attributed to the "stirring effect"—wave action and turbulence generated by the robot's propulsion disturbing the oxygen-depleted boundary layer at the electrode membrane (Shete et al., 2024; Flores-Iwasaki et al., 2025). Despite this, the absolute RMSE values remained below 0.18 mg/L, well within the operationally acceptable range given the 3.0 mg/L hypoxia threshold for Nile Tilapia (Li et al., 2018), consistent with Vasudevan & Baskaran (2021).

Significance of Sensor Differences

A paired t-test was performed for each parameter across each pond to determine whether differences between the Project D.E.P.T.H. sensor readings and the calibrated commercial reference were statistically significant (critical t = ±2.131 for df = 15, α = 0.05). Results are consolidated in Table 5.

Table 5. *Paired t-Test Summary for Sensor Comparison (df = 15, α = 0.05, Critical t = ±2.131)*

Parameter	**Pond 1 (t)**	**Pond 2 (t)**	**Pond 3 (t)**
pH	-13.82*	-21.91*	-21.91*
Temperature (°C)	-3.87*	0.43	0.36
DO (mg/L)	-0.91	2.81*	-0.39

Note. * denotes statistical significance (|t| > 2.131).

Hypothesis H02—that there would be no significant difference between robot and commercial sensor readings—was partially rejected. For pH, all three ponds yielded statistically significant results, indicating a systematic underestimation, which is predictable, correctable with a calibration offset, and operationally negligible (Ibrahim et al., 2016; Bouillod et al., 2022). For temperature, only Pond 1 was significant. For DO, only Pond 2 was significant. The selective significance reflects the greater variability in outdoor conditions rather than a persistent sensor deficiency. Overall, the absolute magnitudes of these differences are well below aquaculturally critical thresholds.

Operational Reliability

The operational reliability of Project D.E.P.T.H. was evaluated using the CV applied to MATE and mission completion time across the three trials. Results are presented in Table 6.

Table 6. *Operational Reliability Summary Across Three Trials*

Metric	Trial 1	Trial 2	Trial 3	Mean	CV (%)
MATE (cm)	2.0187	3.1688	2.3375	2.51	50.67
Mission Time (min)	30.2	31.1	30.5	30.60	2.94

Note. CV = Coefficient of Variation. High reliability is defined as CV < 10%.

Mission completion time demonstrated high reliability (CV = 2.94%), well below the 10% threshold, indicating that the robot's propulsion and timing-based navigation consistently executes its programmed trajectory within a predictable time window (Davis et al., 2023). However, the overall MATE metric yielded a CV of 50.67%, exceeding the 10% benchmark, primarily attributed to differential environmental conditions across the three ponds. Hypothesis H03 was therefore rejected for the MATE metric. Importantly, even in the worst case, the terminal error of 3.1688 cm represents a minor spatial displacement relative to the 6m × 6m pond, and mission time remained highly consistent. Future iterations should explore hybrid navigation approaches incorporating ultrasonic boundary detection or periodic GPS corrections to reduce cross-trial ATE variability (Salkar et al., 2024).

Visual Interpolation Accuracy

The spatial mapping capability of the web-based dashboard was assessed through a ground truth validation process. Thirteen blind-spot coordinates per pond were selected, yielding 39 total validation points per parameter.

Table 7. *Visual Interpolation Accuracy Summary (Ground Truth Validation, n = 13 per Pond)*

Parameter / Pond	MAE	RMSE	Min AE	Max AE
DO (mg/L) — Pond 1	0.1323	0.1545	0.0000	0.2500
DO (mg/L) — Pond 2	0.1669	0.1893	0.0000	0.2900
DO (mg/L) — Pond 3	0.1592	0.1834	0.0000	0.2700
pH — Pond 1	0.1292	0.1567	0.0100	0.2600
pH — Pond 2	0.1123	0.1256	0.0300	0.2400
pH — Pond 3	0.1331	0.1613	0.0100	0.2700
Temperature (°C) — Pond 1	0.1569	0.1760	0.0300	0.2700
Temperature (°C) — Pond 2	0.1885	0.2076	0.0200	0.2900
Temperature (°C) — Pond 3	0.1423	0.1683	0.0200	0.2900

Note. MAE = Mean Absolute Error; RMSE = Root Mean Square Error. Units match parameter units.

The DO parameter yielded MAE values of 0.1323, 0.1669, and 0.1592 mg/L across the three ponds. Minimum absolute errors of zero confirm that bilinear interpolation achieved near-perfect prediction where the spatial DO gradient was smooth. The highest errors in Pond 2 are consistent with greater surface wind and photosynthetic activity creating steeper dissolved oxygen gradients. Even the worst-case error of 0.29 mg/L represents only 9.7% relative error at the 3.0 mg/L hypoxia threshold—an operationally acceptable margin (Li et al., 2018).

The pH parameter produced the lowest and most consistent interpolation errors (MAE range: 0.1123–0.1331), reflecting the inherently gradual and continuous variation of pH in small freshwater ponds due to the carbonate buffering system. The maximum absolute error of 0.27 pH units represents only 13.5% of the optimal growth range (pH 6.5–8.5) for Nile Tilapia (FAO, 2025b), making it operationally negligible.

Temperature recorded the highest overall interpolation errors, with Pond 2 yielding the maximum RMSE of 0.2076°C. This is physically consistent with the nature of heat distribution in outdoor ponds, where localized solar radiation and wind-driven mixing can create fine-scale thermal gradients not fully captured by a 16-point sampling grid. The maximum absolute temperature error of 0.29°C is less than 0.8% of the total tolerable temperature range for Nile Tilapia, and therefore operationally inconsequential (FAO, 2025b). These results confirm that bilinear interpolation is well-matched to the physical characteristics of small-scale tilapia ponds and is reliable enough to support early hypoxia detection (Gomez et al., 2021).

User Experience

The usability of the web-based dashboard was evaluated using the SUS administered to the three tilapia farmers. Individual question scores and summary SUS scores are presented in Tables 8 and 9.

Table 9. *Summary of SUS Scores and Usability Classification*

Respondent	SUS Score	Grade	Adjective Rating	Acceptability
Farmer A	77.5	D	OK	Acceptable
Farmer B	95.0	A	Best Imaginable	Acceptable
Farmer C	65.0	F	Poor	Marginal
Group Mean	79.17	D	OK	Acceptable

Table 8. *Individual SUS Question Scores (1–5 Scale) for Each Farmer*

Question Summary	Farmer A	Farmer B	Farmer C
Would like to use frequently	4	5	4
Unnecessarily complex	2	1	2
Easy to use	4	5	4
Would need technical support	2	1	3
Functions well integrated	5	5	4
Too much inconsistency	1	1	2
Most people would learn quickly	4	5	3
Cumbersome to use	2	1	2
Felt confident using it	4	5	3
Needed to learn many things first	2	1	3

The three farmers produced SUS scores of 77.5, 95.0, and 65.0, yielding a group mean SUS score of 79.17. This mean score falls within the "Acceptable" range (≥ 68) as defined by Brooke (1996), corresponding to an adjective rating of "OK." Hypothesis H04 was rejected: the web-based dashboard did achieve the threshold for Acceptable Usability. Farmer B's near-perfect score of 95.0 likely reflects greater prior familiarity with web-based applications. Farmer C's marginal score of 65.0 suggests that additional farmer onboarding—such as a brief tutorial or simplified dashboard mode—could elevate future usability scores. These findings are consistent with Rosnita et al. (2024) and Essamlali et al. (2024), who noted that end-user usability of IoT aquaculture dashboards is significantly enhanced by simplified visual interfaces and user training.

CONCLUSIONS

Project D.E.P.T.H. successfully demonstrated the technical feasibility of an autonomous surface vehicle for spatiotemporal water quality mapping in small-scale Nile Tilapia aquaculture ponds. The system's dead reckoning navigation, while prone to predictable cumulative drift, consistently delivered terminal errors well within the tolerable range for the pond dimensions tested, validating the core navigation concept for this application scale.

The sensors proved capable of reliably detecting water quality conditions relevant to hypoxia management. The pH and temperature sensors performed with high accuracy and negligible error, while the DO sensor—the most critical parameter for this study's objective—demonstrated sufficient accuracy for meaningful hypoxia detection across all ponds, despite elevated variability in exposed outdoor environments. The statistically significant biases detected for some parameters represent systematic, correctable offsets rather than erratic sensor failures, and their practical impact on aquaculture management decisions is minimal. The overall evidence leads to the conclusion that Project D.E.P.T.H.'s sensor platform is a viable alternative to professional-grade handheld meters for routine water quality profiling in tilapia ponds.

The web-based dashboard succeeded in translating raw sensor data into an accessible and actionable spatial visualization for end-users. The heatmap interpolation accuracy was consistent and maximum errors remained within safe operational margins. Critically, the SUS evaluation demonstrated that the platform achieved acceptable usability among real-world tilapia farmers, confirming that the system bridges the gap between data collection and farm-level decision-making. Taken collectively, this study demonstrates that an integrated ASV-IoT monitoring system can be assembled from commercially available components to provide tilapia farmers with a practical and meaningful upgrade to their water quality management capabilities, with the potential to significantly reduce the frequency and severity of hypoxia-induced fish kills.

RECOMMENDATIONS

To address the cross-trial variability in terminal navigation error (CV = 50.67%), future iterations of Project D.E.P.T.H. should incorporate a hybrid navigation approach, such as a low-cost ultrasonic distance sensor array or a single-fix GPS module, to provide periodic boundary or position corrections during the mission. This would directly target the operational reliability gap and improve the system's deployability across ponds with varying environmental conditions (Salkar et al., 2024).

To reduce the systematic bias in pH readings and the elevated DO variability in outdoor deployments, future studies should implement an automated in-situ calibration protocol and explore physical DO probe shielding or water pumping mechanisms to isolate the galvanic electrode from wave-induced disturbances. Testing the system across a wider variety of pond environments—including larger commercial ponds and ponds with denser fish stocking—would also provide a more robust empirical basis for sensor validation. Finally, to improve dashboard usability for farmers with limited digital literacy, future deployments should include a structured onboarding session with a simplified tutorial, a mobile-optimized interface with multilingual support (including Filipino/Bisaya), and one-tap alert notifications. A larger-scale SUS evaluation with a broader pool of farmer respondents is also recommended to yield statistically robust usability conclusions and guide iterative interface improvements.

ACKNOWLEDGMENT

The researchers wish to express their sincerest gratitude to all those who extended their invaluable assistance, and support throughout the completion of this study. They are deeply grateful to their research adviser, Ariel V. Fabrigas, for the expertise in software development and constructive feedback offered at every stage of this project, and to their Capstone teacher, Engr. Shoji-an D. Daradal, for the unwavering mentorship and encouragement. They also extend their gratitude to their Practical Research II teacher, Janis Marie P. Alagon, for her continuous guidance; to Nolan V. Alvear and the members of the Research Presentation Panel for their expert evaluations and insightful recommendations; and to the three tilapia farm owners whose cooperation and patience during the deployment trials were indispensable. Deepest thanks are due to the families of the researchers for their constant moral support, and above all, to God Almighty, for wisdom, strength, and guidance throughout this journey.

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